

How does this argument become closer to the base case? The range halves in size for each recursive call, so it eventually becomes one item long.

Examine the following `binarySearch()` function in our *binarySearch.py* program, which locates a value, `needle`, in a sorted list of values, `haystack`:

Python

```
def binarySearch(needle, haystack, left=None, right=None):
    # By default, `left` and `right` are all of `haystack`:
    if left is None:
        left = 0 # `left` defaults to the 0 index.
    if right is None:
        right = len(haystack) - 1 # `right` defaults to the last index.

    print('Searching:', haystack[left:right + 1])

    if left > right: # BASE CASE
        return None # The `needle` is not in `haystack`.

    mid = (left + right) // 2
    if needle == haystack[mid]: # BASE CASE
        return mid # The `needle` has been found in `haystack`
    elif needle < haystack[mid]: # RECURSIVE CASE
        return binarySearch(needle, haystack, left, mid - 1)
    elif needle > haystack[mid]: # RECURSIVE CASE
        return binarySearch(needle, haystack, mid + 1, right)

print(binarySearch(13, [1, 4, 8, 11, 13, 16, 19, 19]))
```

The *binarySearch.html* program has this JavaScript equivalent:

JavaScript

```
<script type="text/javascript">
function binarySearch(needle, haystack, left, right) {
    // By default, `left` and `right` are all of `haystack`:
    if (left === undefined) {
        left = 0; // `left` defaults to the 0 index.
    }
    if (right === undefined) {
        right = haystack.length - 1; // `right` defaults to the last index.
    }
}
```